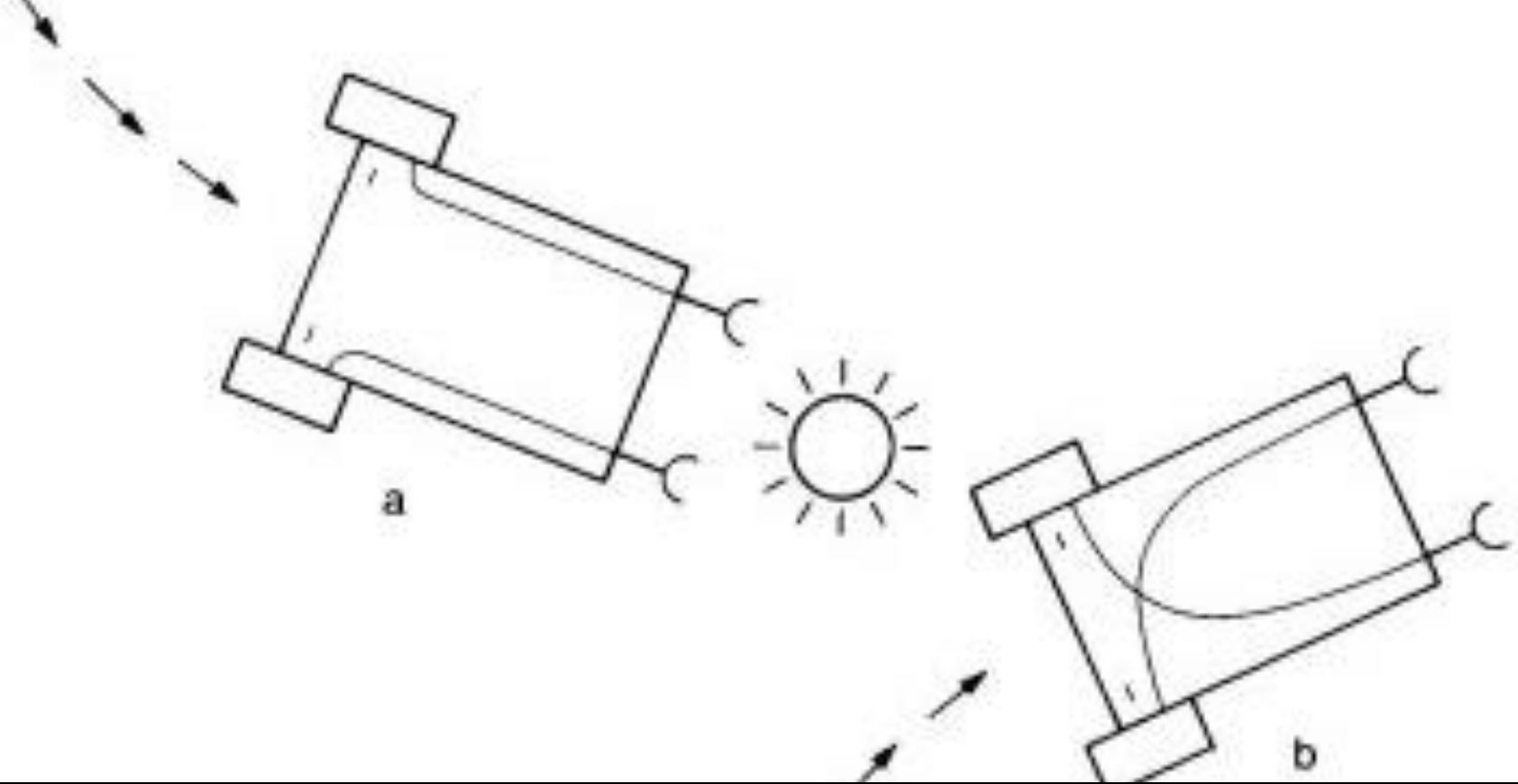




Introduction to Robotics – Part I – a historical perspective

Kasper Støy (ksty@itu.dk)

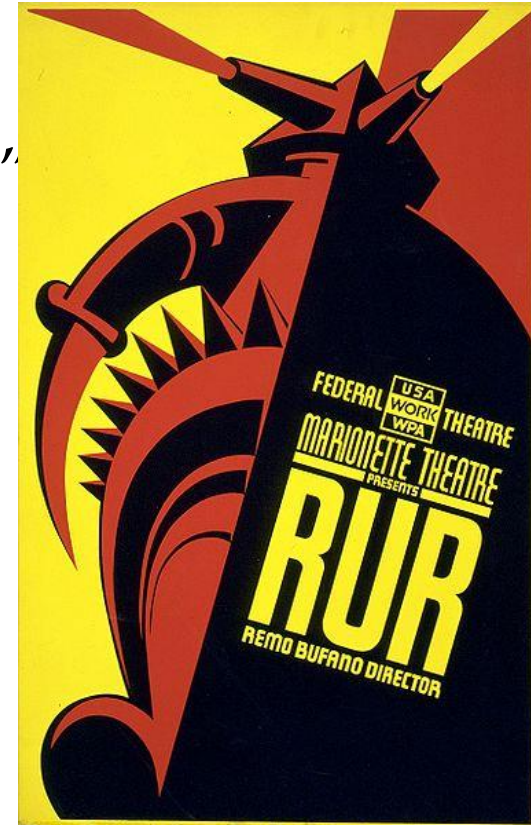
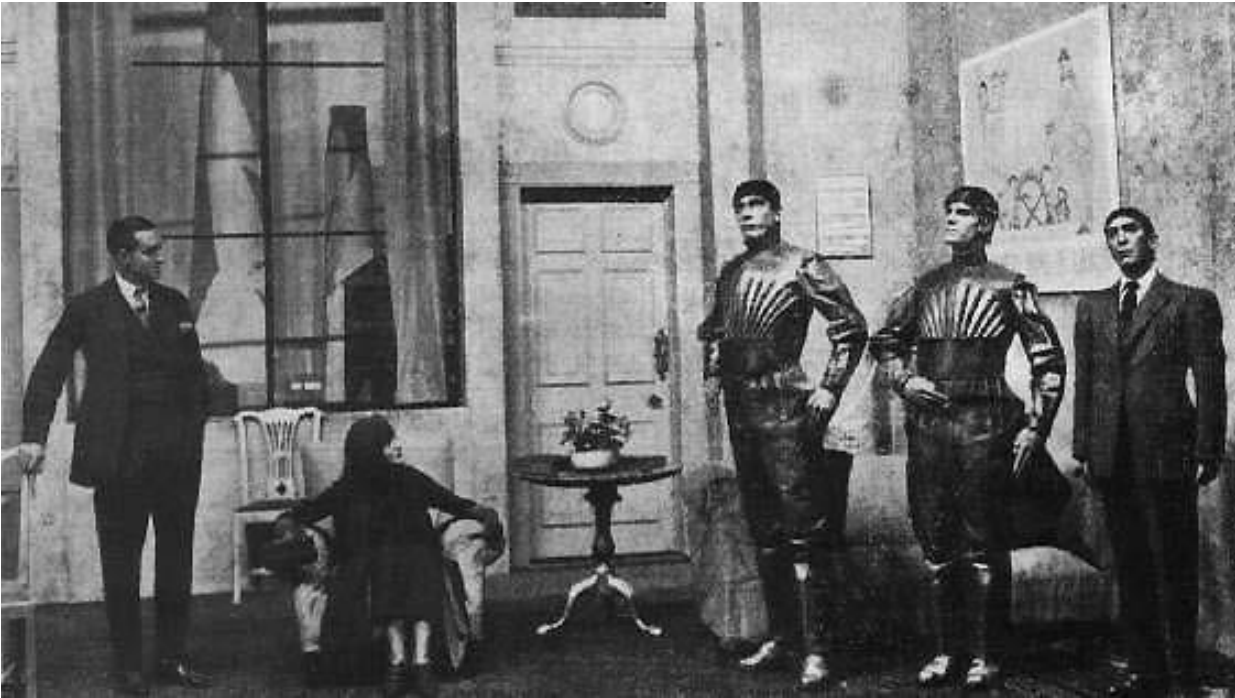
ORIGIN OF ROBOTS

A theater play!

Karel og Josef Capek:

Rossum's Universal Robots (1920)

- "They'd make fine university professors."



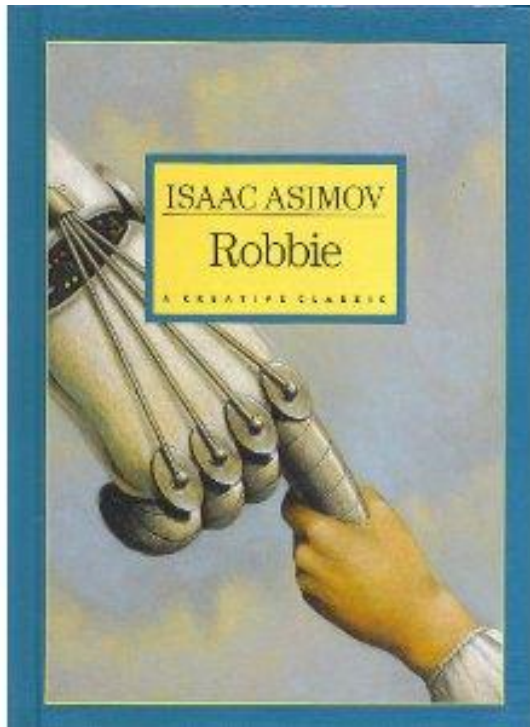
New York, 1939

Books!

Isaac Asimov:

Robbie (1939),

Runaround (1942)



1965

Asimov's laws of robotics

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

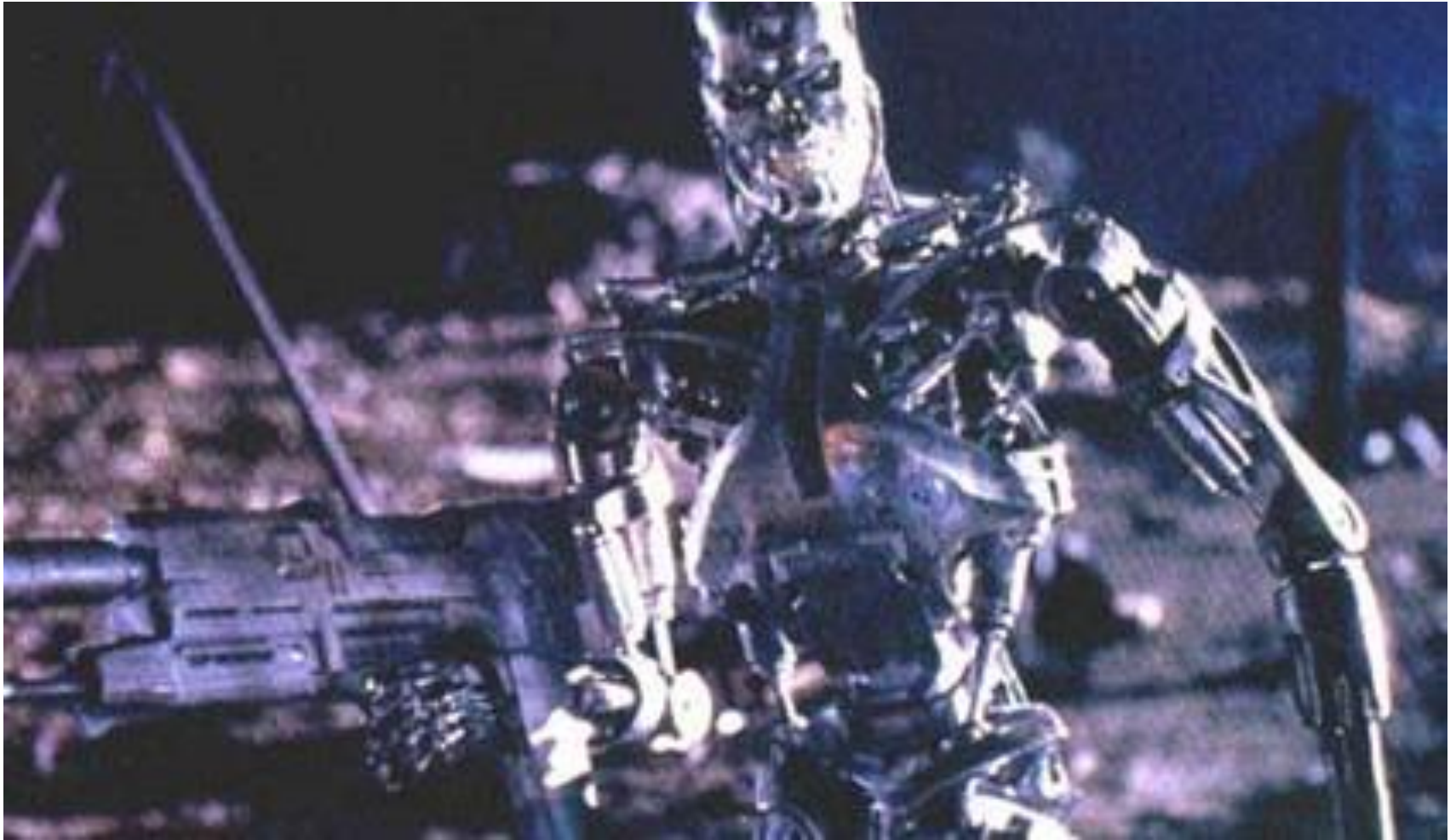
Robots go to Hollywood



Versatile human-like robots (who want to be human)?
Powerful but (fortunately) limited machines?
Funny? Pitiful?
Explores boundaries for what it means to be human

Movie roles of robots

A threat!!!



THE ROBOTS OF REALITY

Robots from the Industrial revolution

Power loom (1733)

Steam engine (1760-)

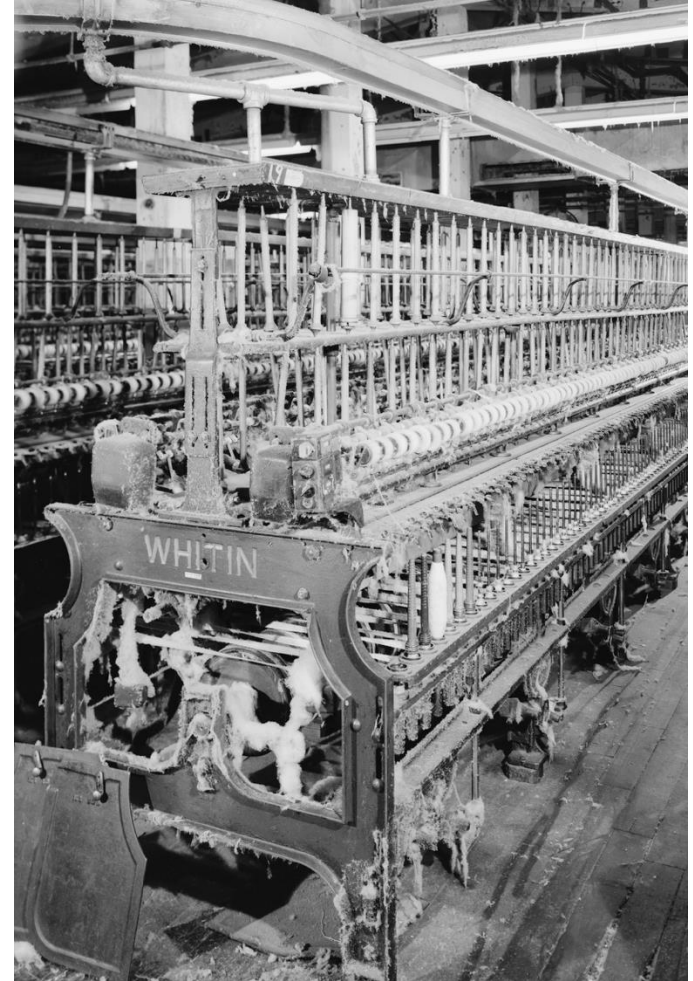
Worm gear (1774)

Mass Production (1908)

Henry Ford's car factory



Source: Wikimedia Commons

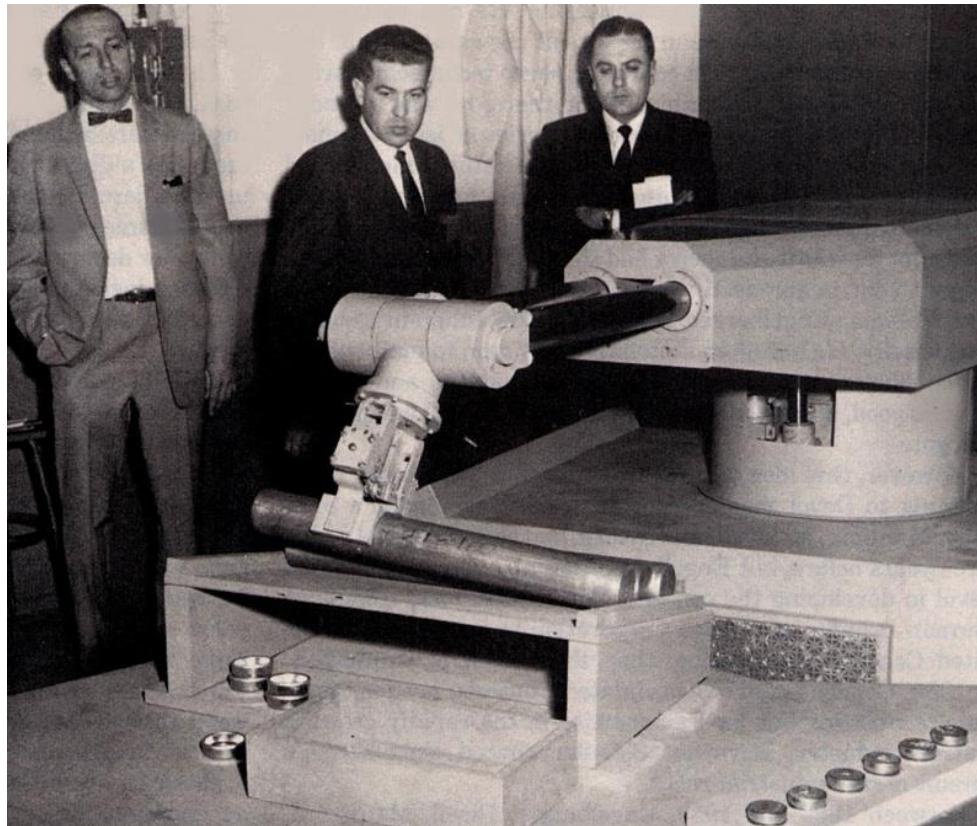


Source: U.S. Library of Congress, Prints and Photographs Division, "Built in America" Collection

Origin of Robots-1954

George Devol - Unimation Inc.

Robots introduced at General Motors in 1961



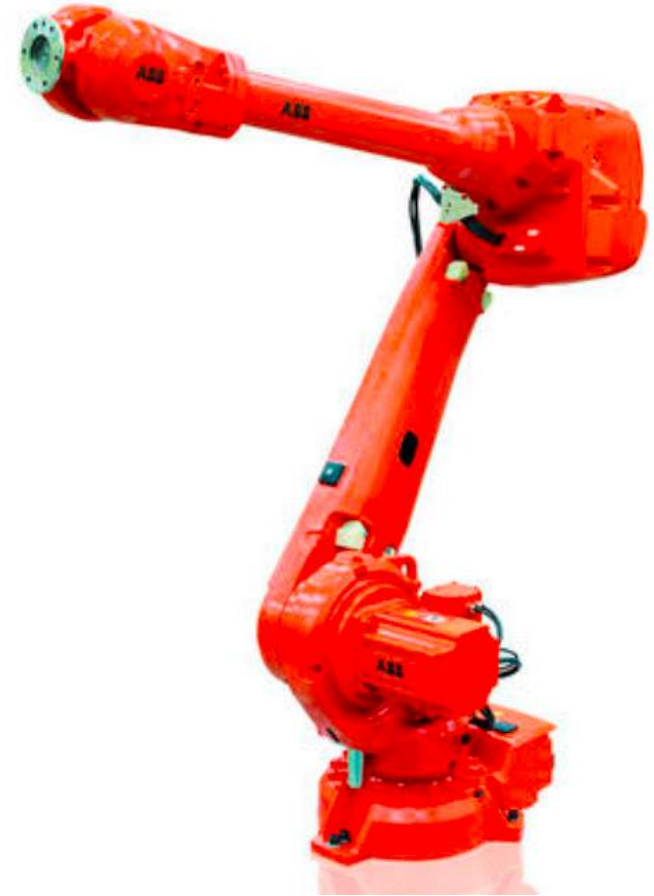
Robot technology in year 2005

Remaining tasks

- Tough for robots
- Not profitable to solve with robots

Robot kr. 300.000

System integration kr. 1.000.000

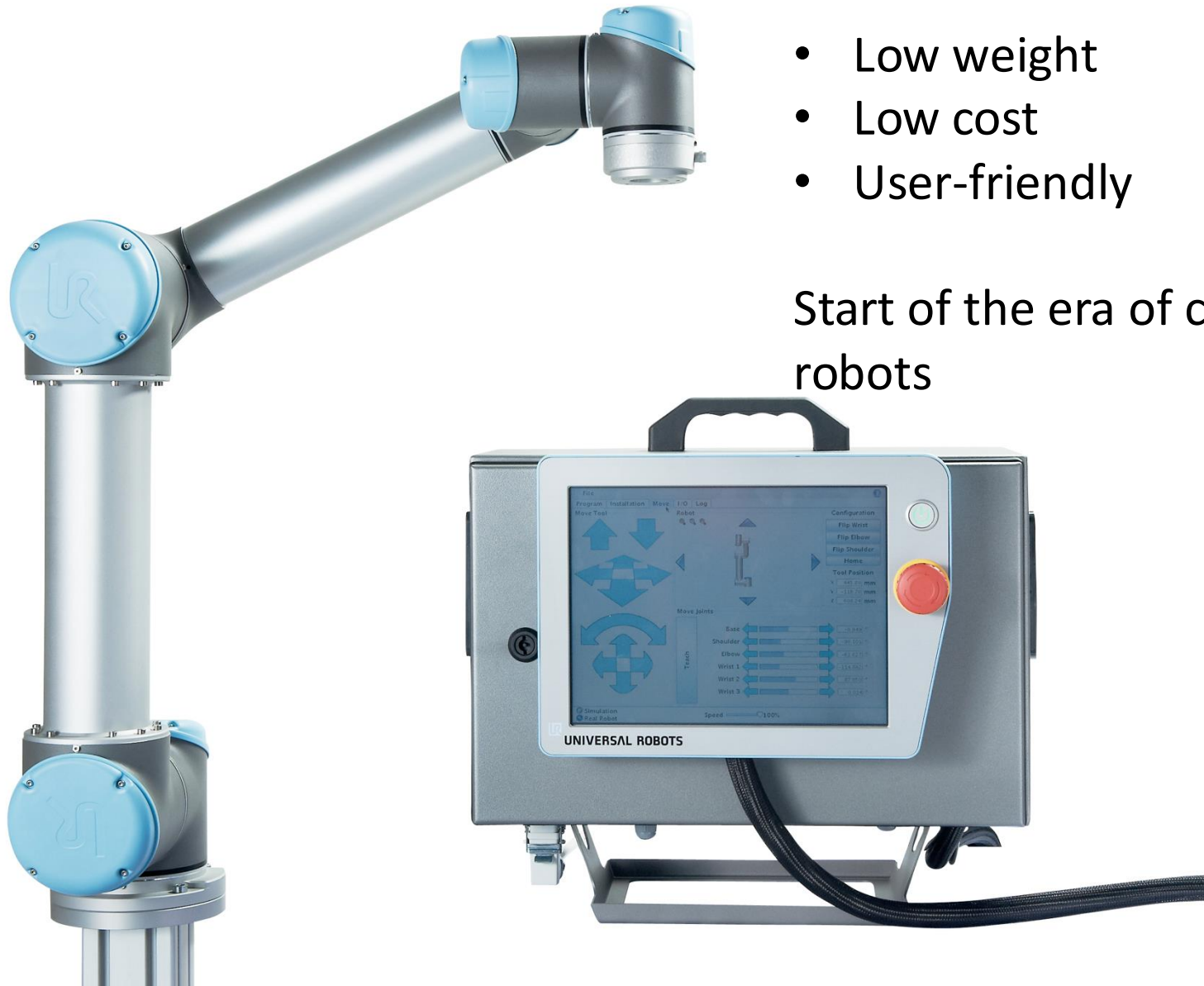


Universal Robots A/S - 2005

The focus of our robot arm UR5

- Low weight
- Low cost
- User-friendly

Start of the era of collaborative robots



Development of the robot (~2006)



Can we glimpse the end of the industrial revolution?

Inertia (8% of manufacturing companies have automation)
Remaining tasks difficult for robots



ROBOTS AND ARTIFICIAL INTELLIGENCE

The idea of artificial brains

Wiener, Turing, Neumann, McCulloch

The fathers of computer science

"You can make computers that can simulate brain functions" (1940s)



Alan Turing



John von Neumann

Thinking robots



Shakey - SRI – 1966 - 1972

Illustrative game

A robot consists of actuators, sensor, and control

Hand (blind) (motor)

- Do what the brain tells you
- Stop if the eyes say "Stop!"

Eyes (sensor)

- If the hand is in danger it "Stop!"
- Answers to questions from the brain – only yes/no and geometric questions are allowed

Brain (blind) (control)

Build a tower of three cups



Tortoise - Grey Walters – 1948 - 1951

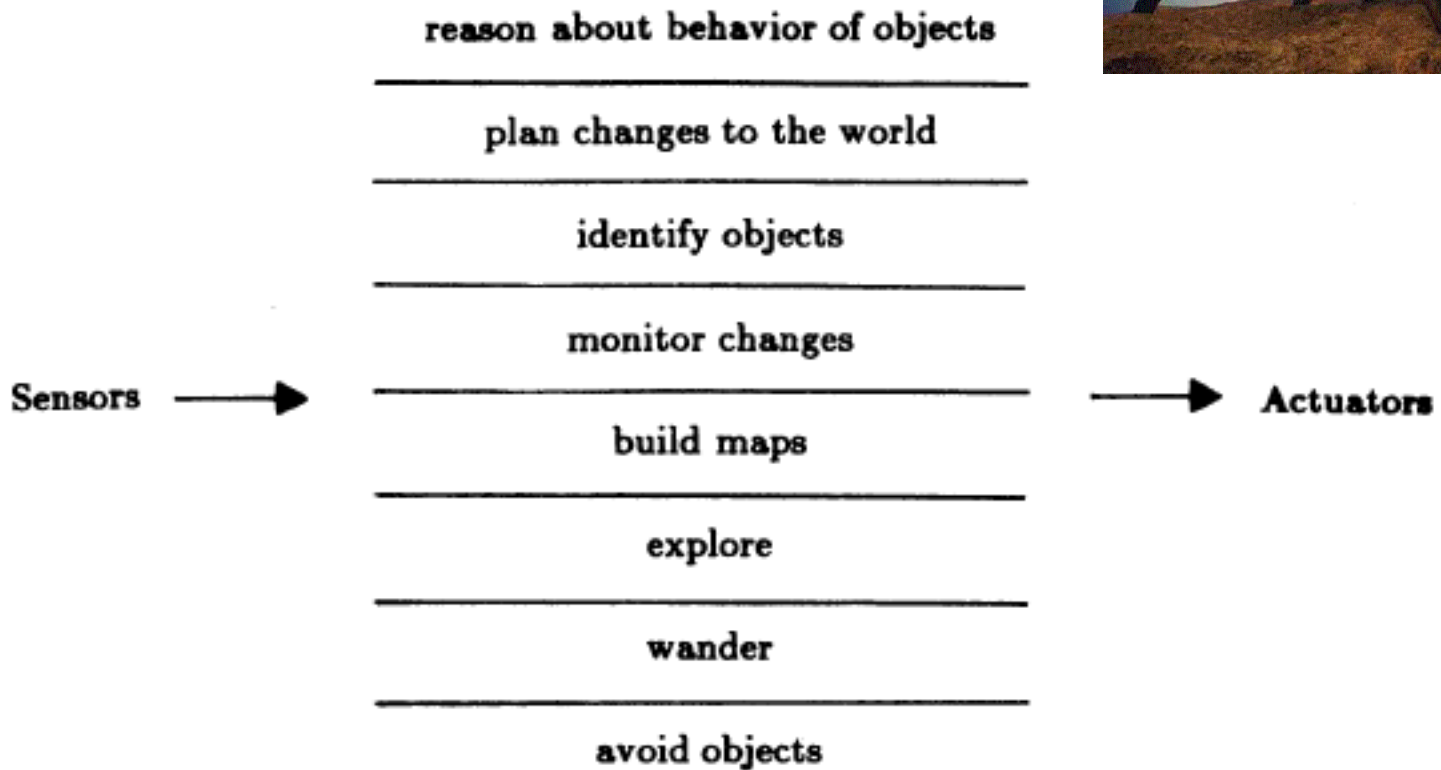
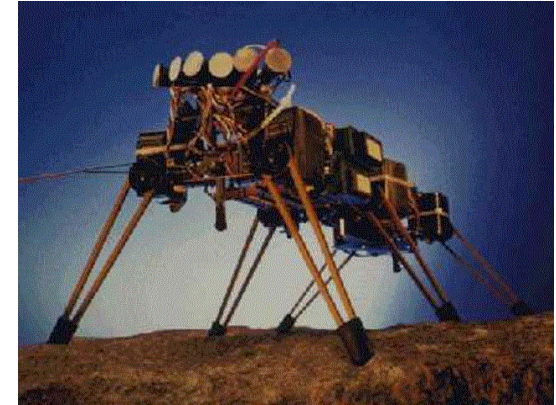
Behavior-based robots



Ghengis – Rodney Brooks, MIT, 1989

A Robust Layered Control System for a Mobile Robot

1986



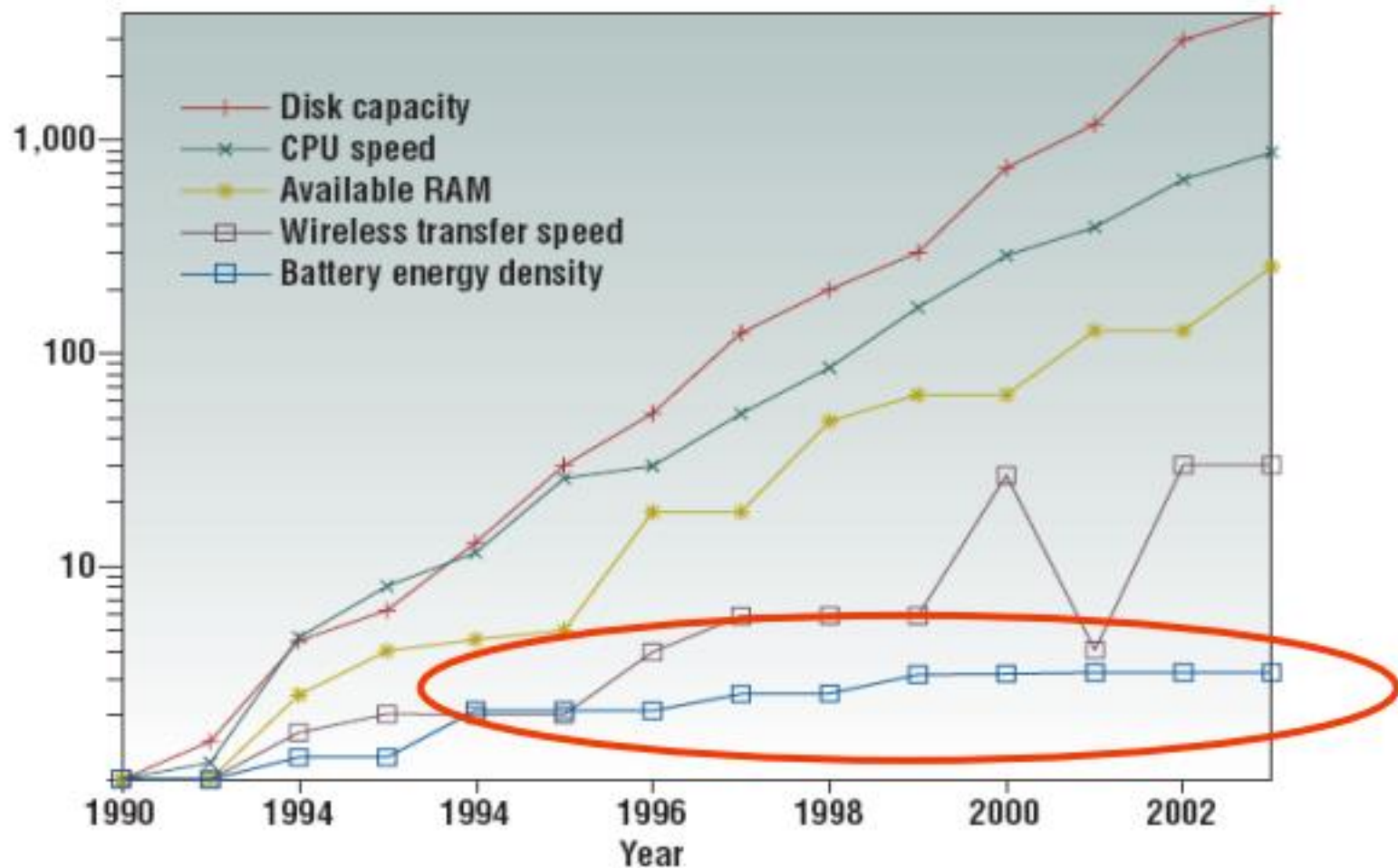
Embodied Artificial Intelligence



Passive-dynamic walking robot, Steve Collins et al., Cornell, 2005

ONE LIMITATION OF ROBOTS

Robot development = Computer development? No



Mobile Computing Improvement – Paradiso, et al. Pervasive Computing, IEEE, 2005.

Robot development = Computer development? No

Robots have to obey the laws of Nature:

$$\text{Potential energy} = \text{mass} * \text{gravity} * \text{height}$$

Example:

$$\text{Energy to lift a robot} = 100\text{kg} * 10 \text{ m/s}^2 * 1\text{m} = 1000 \text{ joule}$$

$$\text{Energy for a 1000 lifts} = 1000 * 1000 = 1.000.000 \text{ joule} = 277 \text{ wh}$$

WITHOUT LOSS in ELECTRONICS, ACTUATORS, GEARS, ETC.

Battery capacity development over the last ten years..

Watt hours per kg of battery



Energy is a fundamental
limitation of robots!

EXAMPLES OF ADVANCED ROBOTS

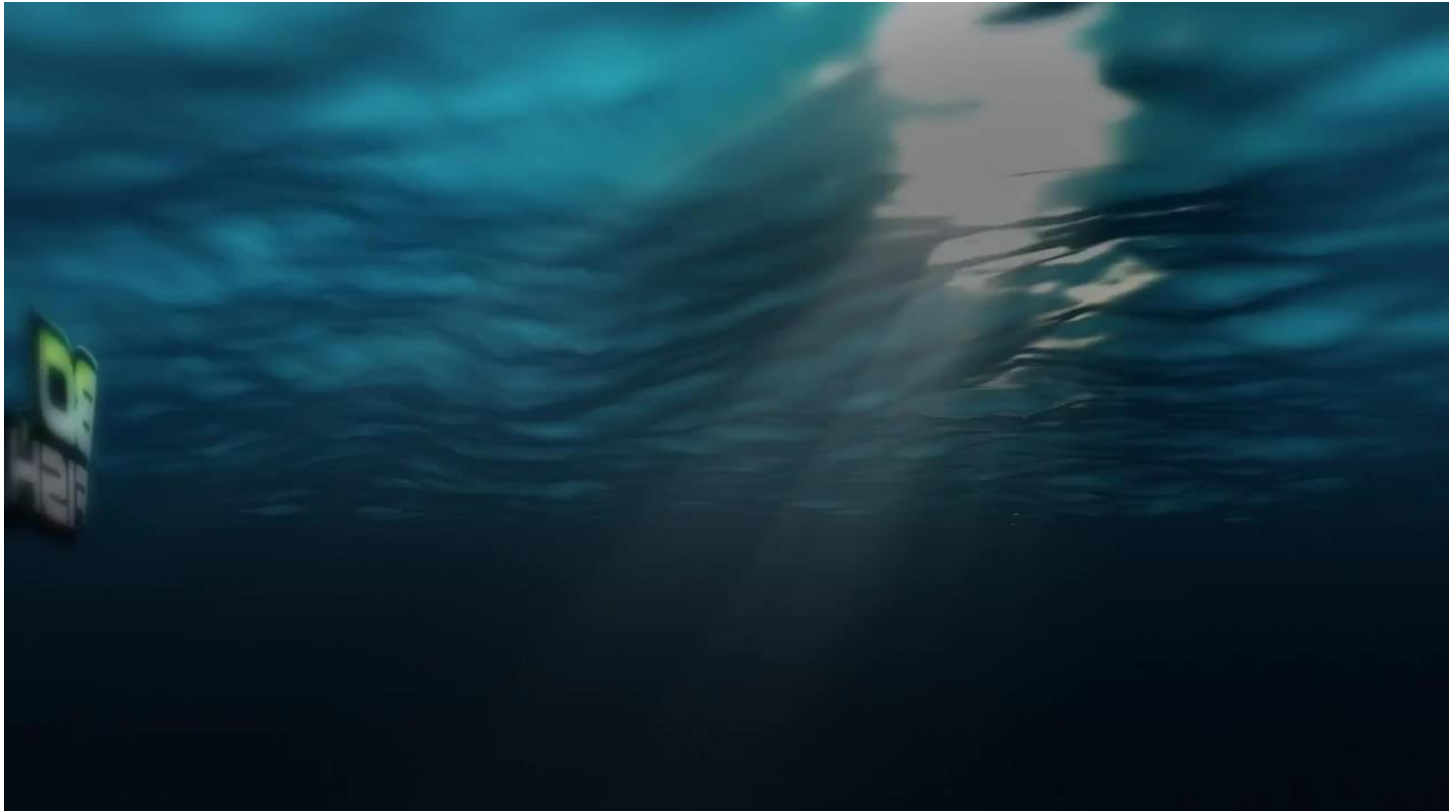
Mobile Industrial Robots - Odense

Flow Robotics A/S

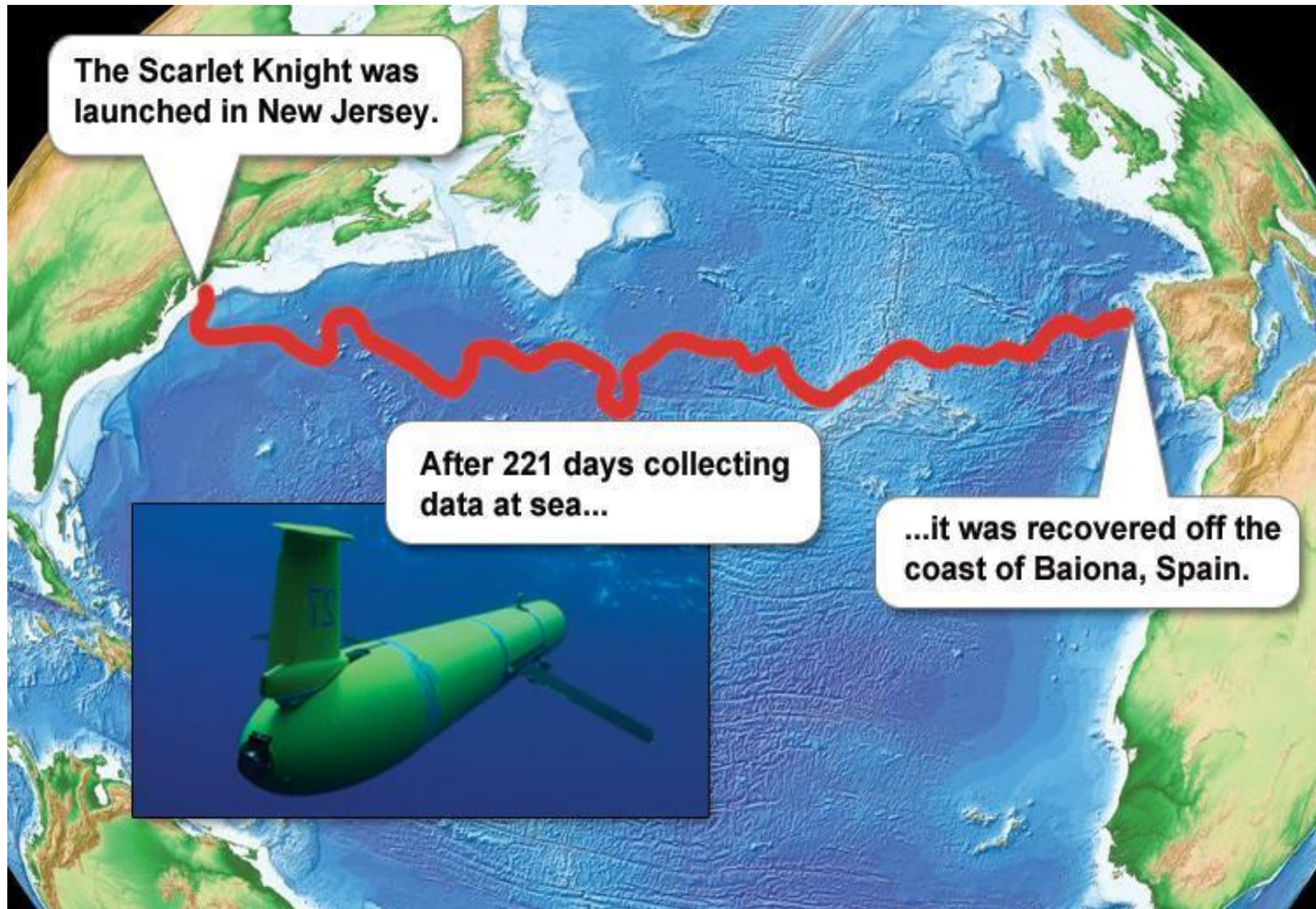


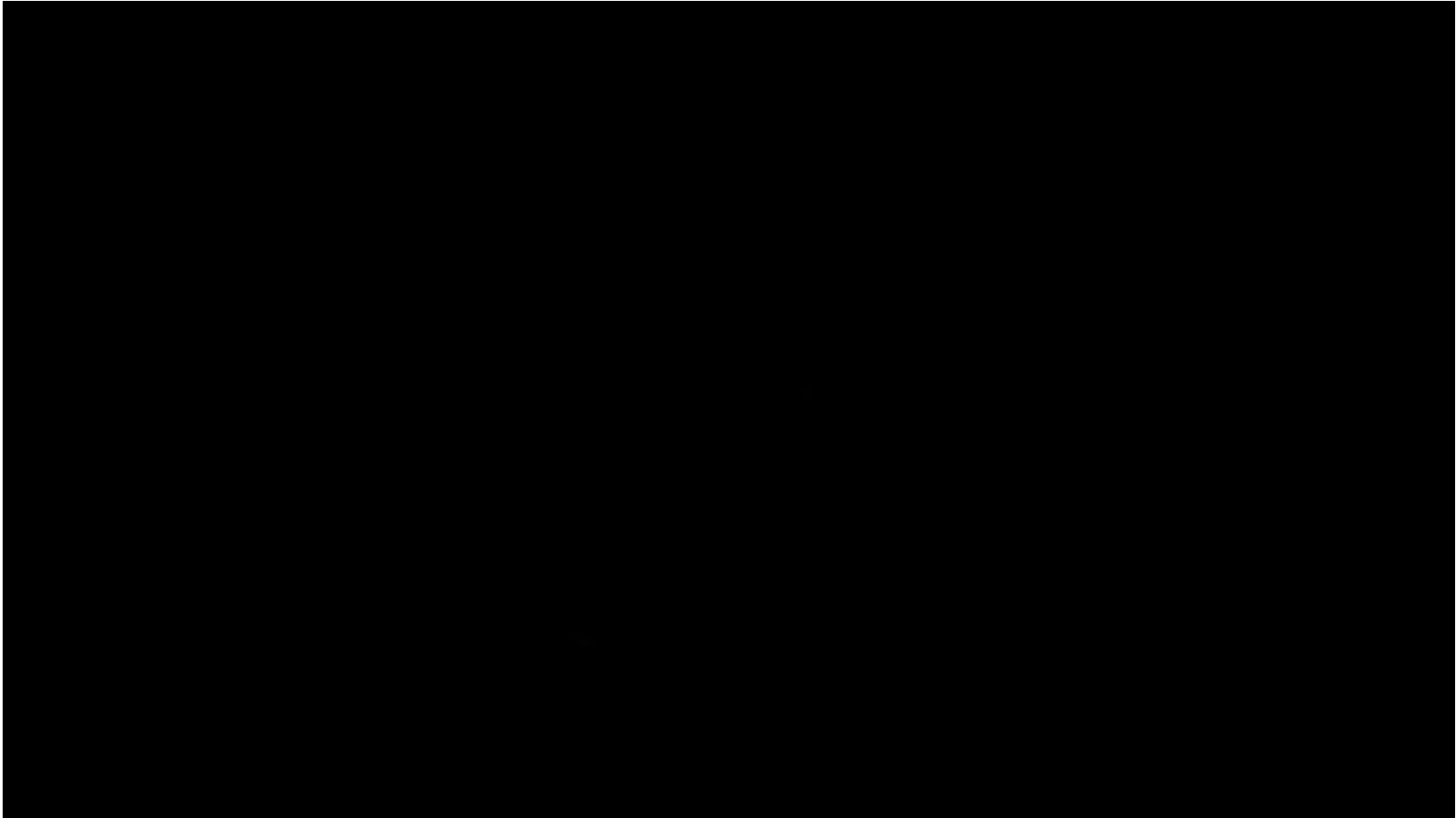
Underwater robotics

The world's fastest fishing robot – iSplash-II
Length 32cm-speed 3.7 m/s (13.3 km/h)



Longest robot journey: Scarlet Knight



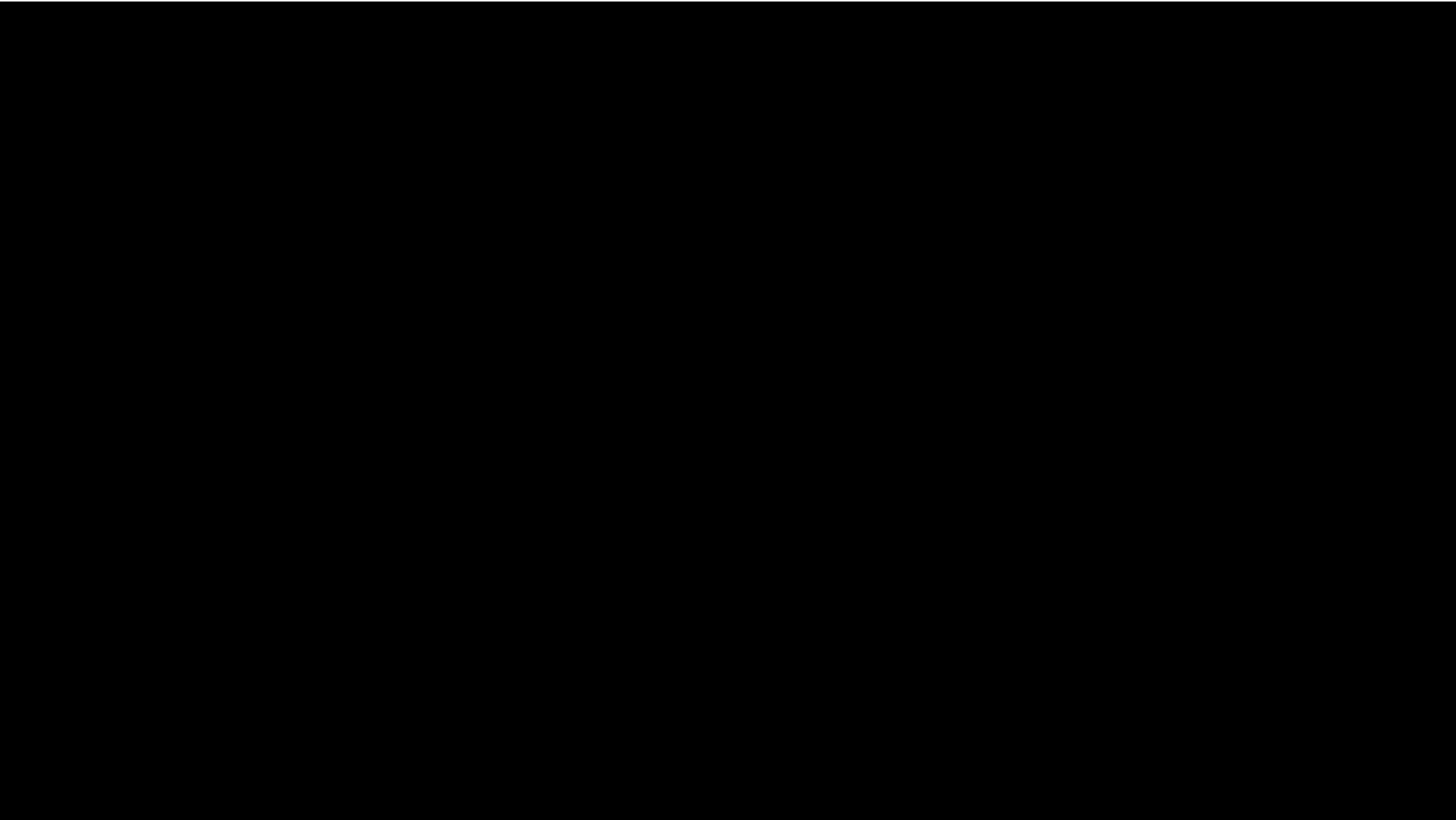


UPenn

On land: walking robots



On land in ..



On land: self-driving cars



On humans



And many more..

- Service Robots (Vacuuming, mowing, gutter cleaning, floor washing, hedge trimming,...)
- Mars rover
- Search and Rescue
- Toys
- Education (Teaching)
-

- Origin of robots in automotive industry and science fiction
- From hierarchical control to embodied A.I. (more to come!)
- Many potential applications, but robots are not likely to take-off, literally speaking